

Formats & Terminology Guide

A structured reference to NIfTI, GIFTI, CIFTI, grayordinates, cortical surfaces, preprocessing, parcellation, and functional connectivity.

Designed for: learning the Human Connectome Project (HCP) Young Adult data and getting started with Connectome Workbench.

Learning path: MRI basics → file formats → spatial representations → HCP pipelines → connectivity analysis

How to use this guide

Use this document as a reference while browsing HCP files. For every file, ask three questions: (1) What is the spatial unit—voxel, vertex, grayordinate, or parcel? (2) What does the value represent—BOLD signal, thickness, activation, or a label? (3) What are the dimensions—volume, location $\times$ time, or region $\times$ region?

Important nuance: exact filenames, available structures, dimensions, and preprocessing products vary by HCP release and data package. Always inspect the file metadata and documentation for the specific file you downloaded.

1. MRI foundations

Voxel and volume

A voxel is a three-dimensional element of a sampled image, analogous to a pixel with depth. A volumetric MRI may have dimensions $X \times Y \times Z$. A 4D fMRI image adds time: $X \times Y \times Z \times T$. Each voxel can therefore have a time series across T acquired volumes.

```
3D structural image:  X  $\times$  Y  $\times$  Z
4D fMRI image:       X  $\times$  Y  $\times$  Z  $\times$  T
One voxel's time series: v(1), v(2), ..., v(T)
```

BOLD, T1, TR, and TE

Term	Meaning	Why it matters
T1-weighted MRI	Structural contrast useful for anatomy and tissue boundaries.	Supports anatomical reconstruction and segmentation.
BOLD fMRI	Blood-oxygen-level-dependent signal; an indirect, vascularly mediated correlate of neural activity.	Used for resting-state and task fMRI.
TR	Repetition time: time between repeated volume acquisitions in a sequence.	Sets temporal sampling; check metadata rather than assuming a value.
TE	Echo time: time from excitation to signal readout.	Affects signal contrast and BOLD sensitivity.
Multiband / SMS	Acquires multiple slices simultaneously.	Enables faster acquisitions and shorter TRs.

Remember: BOLD is not a direct recording of neuronal firing. It reflects changes in blood oxygenation and hemodynamics associated with neural and non-neural processes.

2. NIfTI: volumetric data

NIfTI (commonly .nii or .nii.gz) is a neuroimaging format widely used for 3D and 4D image volumes. The header describes dimensions, voxel sizes, datatype, and spatial transforms/orientation; the file also stores the image data.

Typical examples

```
T1w.nii.gz           → 3D anatomical volume
rest_bold.nii.gz     → often a 4D BOLD volume
brain_mask.nii.gz    → volume mask
```

A NIfTI coordinate such as voxel index (i, j, k) is not automatically the same as a physical coordinate in millimetres. Affine transforms in the header relate voxel indices to a spatial reference. Check orientation and transforms when comparing images.

Useful Python inspection

```
import nibabel as nib
img = nib.load('image.nii.gz')
print(img.shape)          # dimensions
print(img.affine)         # voxel-to-world transform
print(img.header.get_zooms())
```

A compressed .nii.gz file is still NIfTI; gzip compression changes storage size, not the conceptual image type.

3. GIFTI: surface-based data

GIFTI (.gii) is commonly used for cortical surface geometry, vertex-wise metrics, and labels. A surface is a mesh made of vertices connected into triangular faces.

Geometry vs metric vs label

Data type	Contains	Example
Surface geometry	Vertex coordinates plus triangle topology.	*.surf.gii
Shape/metric	A value at each vertex.	*.shape.gii; thickness or curvature
Functional metric	Vertex-wise functional/statistical values.	activation or mapped BOLD values
Label	Categorical region assignment per vertex.	*.label.gii

Common HCP surface names

Name	Interpretation
white	Approximate gray-matter/white-matter boundary.
pial	Approximate outer cortical boundary adjacent to CSF.
midthickness	Surface between white and pial surfaces; commonly used for analyses.
inflated / very_inflated	Visualization surfaces that reduce the visual impact of folds.
sphere	Spherical representation often used for surface registration.

A surface file alone describes geometry; to colour it by thickness or activation, load a matching metric or label file. The geometry and metric must have compatible vertex counts and spaces.

4. CIFTI and grayordinates

CIFTI (Connectivity Informatics Technology Initiative) is a family of data structures designed to represent brain data across cortical surfaces and selected subcortical or other gray-matter locations. Many CIFTI filenames end in .nii, but they are not ordinary 3D/4D NIfTI volumes: CIFTI stores mappings that explain what each data element represents.

What is a grayordinate?

A grayordinate is one element of a combined gray-matter representation. Depending on the structure, it may correspond to a cortical surface vertex or a subcortical voxel. A dense CIFTI therefore combines surface-based cortical locations with volume-based locations in one mapped data structure.

```
Cortical grayordinates: surface vertices
Subcortical grayordinates: selected voxels
Combined mapping: grayordinates × measurements/time
```

The often-mentioned HCP “91k” representation is a common dense grayordinate space (frequently around 91,282 locations), not simply 91,000 volume voxels. The precise composition depends on the

5. CIFTI filename cheat sheet

Extension	Meaning	Typical dimensions / use
.dtseries.nii	Dense time series	Grayordinate × time; often preprocessed fMRI.
.dscalar.nii	Dense scalar	One or more scalar maps over grayordinates, e.g. activation/statistics.
.dlabel.nii	Dense labels	Categorical labels over grayordinates, e.g. a parcellation.
.ptseries.nii	Parcellated time series	Parcel × time; useful for connectivity.
.pscalar.nii	Parcellated scalar	Scalar values per parcel.

The prefix is a useful clue, not a substitute for checking the file's actual dimensions and metadata. A CIFTI file may have more than one map or an additional mapped dimension.

6. Surface space, registration, and resolution

Native and standard space

Native space refers to an individual's own anatomical coordinate system. Standard/template space brings data into a common reference so locations can be compared across people. Spatial normalization is the broader process of mapping an individual's data to a reference.

MNI and fs_LR

Term	Practical meaning
MNI space	A family of standard volumetric reference spaces used for volume-based analyses.
fs_LR	HCP surface-space convention associated with left/right cortical surface registration.
32k_fs_LR	Approximately 32,000 mesh vertices per hemisphere before accounting for excluded/non-cortical locations in a given mapping.
164k_fs_LR	Higher-resolution surface mesh, approximately 164,000 vertices per hemisphere.
Registration	Aligning images or surfaces so corresponding anatomical locations can be compared.
Medial wall	Medial cortical areas or mesh locations excluded from the cortical grayordinate analysis mapping.

Do not treat MNI volume coordinates and fs_LR surface vertices as interchangeable. They describe different representations and require appropriate mapping tools.

7. Atlas, parcellation, ROI, and masks

Atlas and parcellation

An atlas defines a set of brain regions and their labels. Parcellation is the division or assignment of brain locations into parcels, often by applying an atlas. A parcel is a region containing many voxels or vertices.

HCP-MMP1.0 (also known as the Glasser atlas) defines 180 cortical areas per hemisphere, for 360 cortical areas total. It is a cortical atlas; it should not be mistaken for a complete atlas of all subcortical structures.

Related terms

Term	Definition
ROI	Region of interest chosen for a particular analysis; may be anatomical, atlas-based, or functionally defined.
Mask	An image or surface map selecting locations, often with 1 for included and 0 for excluded.
Label map	Categorical assignments such as parcel IDs.
Scalar map	Numeric values such as thickness, activation, or a statistical estimate.

Dense versus parcellated analysis

Dense data: `grayordinate × time`
Parcellation: `many locations → region summaries`
Parcellated data: `parcel × time`
Connectivity: `parcel × parcel`

Parcellation reduces dimensionality and can make analysis easier, but it also averages within-region variation and makes results dependent on the atlas and extraction method.

8. From time series to functional connectivity

A time series is a sequence of measurements over time. In fMRI, a parcel's series is often extracted by summarizing the signal across its included vertices/voxels at each time point.

Parcel A: `a(1), a(2), ..., a(T)`
Parcel B: `b(1), b(2), ..., b(T)`
Functional connectivity example: `corr(a, b)`

Functional connectivity commonly refers to statistical dependence between regional signals, often measured with Pearson correlation. It does not, by itself, establish direct anatomical connections or causality.

Connectivity matrix and connectome

With 360 parcels, a subject's correlation matrix is typically 360×360 . In a graph representation, parcels are nodes and pairwise relationships are edges. The resulting network is a functional connectome.

`parcel time series → pairwise correlations →  $360 \times 360$  matrix`
`nodes = parcels; edges = chosen connectivity values`

Before calculating connectivity, check motion and other confounds, censoring, filtering, standardization, and whether the data are resting-state or task-based. These choices can change the result.

9. HCP preprocessing and task terminology

Term	Meaning
HCP minimal preprocessing	HCP pipeline for processing structural, diffusion, and functional data into standardized products.
Motion correction	Corrects estimated head movement between image volumes; residual motion can still affect results.
Distortion correction	Addresses susceptibility-related spatial distortion using available acquisition information.
Surface mapping	Maps cortical measurements from volume space to a cortical surface.
ICA	Independent Component Analysis; decomposes data into spatial/temporal components.
ICA-FIX	A component-classification denoising approach used in HCP functional processing.
Task / run / condition	Experiment / acquisition run / experimental condition.
Contrast	A statistical comparison, e.g. 2-back > 0-back.
LR / RL	In HCP acquisition labels, commonly indicate left-to-right or right-to-left phase-encoding direction; confirm the specific file metadata.

Resting-state fMRI measures spontaneous fluctuations while a participant is not performing a structured task. Task fMRI records data while participants perform an experimental paradigm. HCP task families include working memory, emotion, gambling, motor, language, social cognition, and relational processing; available files depend on release and subject.

Always distinguish raw, minimally preprocessed, and further denoised/analysis-ready data. A file being provided by HCP does not mean every possible nuisance source has been eliminated.

10. Opening files in Connectome Workbench

Launch the viewer

```
wb_view
```

Open a file from the terminal

```
wb_view /path/to/file.dtseries.nii
wb_view /path/to/surface.surf.gii
wb_view /path/to/volume.nii.gz
```

Alternatively, launch Workbench and choose File → Open File. Use the real path on your computer; the example paths above are placeholders.

What to expect

- A .dtseries.nii file contains dense time-series data. Workbench can display maps on compatible surfaces and show applicable volume structures.
- A .dscalar.nii file typically displays a scalar map; a .dlabel.nii displays categorical labels.
- A .surf.gii file provides surface geometry. Load a compatible metric/label file to colour the surface with data.
- A .nii or .nii.gz file is volumetric. It may appear in a volume view rather than as a cortical surface map.

If it fails

- Check that Workbench is installed and available on PATH: run which wb_view.
- Check the file path and permissions; spaces in paths should be quoted in the terminal.

- Ensure that surface and metric/label files are compatible in hemisphere, mesh resolution, and coordinate space.
- For large files, allow time for loading and check the terminal for error messages.

11. A practical HCP analysis workflow

1. Identify the file and inspect its metadata/dimensions.
2. Confirm subject, task/run, preprocessing level, and space.
3. Visualize the data in Workbench.
4. Select a compatible atlas/parcellation if regional analysis is intended.
5. Extract parcel time series and review quality/confounds.
6. Estimate connectivity (e.g., correlation matrix).
7. Test brain-behavior relationships with appropriate statistics.
8. Validate models across subjects; avoid leakage in ML/XAI.

For decision-making or social-cognition projects, first confirm which task data and behavioral measures are available in the exact HCP release you are using. Resting-state connectivity and task activation answer different questions and should not be treated as equivalent.

12. Quick reference tables

If you see...	Think...	Spatial unit
.nii / .nii.gz	Volumetric image	Voxel
.surf.gii	Surface geometry	Vertex + mesh faces
.shape.gii	Surface metric/shape values	Vertex
.label.gii	Surface labels	Vertex
.dtseries.nii	Dense time series	Grayordinate × time
.dscalar.nii	Dense scalar map	Grayordinate × scalar map
.dlabel.nii	Dense labels	Grayordinate × label
.ptseries.nii	Parcellated time series	Parcel × time
.pscalar.nii	Parcellated scalar	Parcel × scalar map

Three questions to ask of every file

- What is the unit? Voxel, vertex, grayordinate, or parcel?
- What does each value mean? BOLD, thickness, activation/statistic, label, or connectivity?
- What are the dimensions? $X \times Y \times Z$ [$\times$ time], grayordinate $\times$ time, parcel $\times$ time, or parcel $\times$ parcel?

Glossary

Term	Definition
BOLD	Blood-oxygen-level-dependent fMRI signal.
CIFTI	Mapped neuroimaging data structure combining brain models such as cortical surfaces and subcortical voxels.
Connectome	Representation of brain connections as a network.
Cortical surface	Mesh representation of the cerebral cortex.
Dense	Data represented at many individual grayordinates rather than summarized into parcels.
Edge	A relationship between two nodes in a network.
Face	A polygon, usually a triangle, connecting mesh vertices.
Functional connectivity	Statistical relationship between neural-imaging time series.
GIFTI	Format commonly used for surface geometry, metrics, and labels.
Grayordinate	A location in a combined gray-matter mapping; typically a cortical vertex or subcortical voxel.
ICA	Independent Component Analysis.
MNI	Montreal Neurological Institute reference-space family.
Node	A unit in a graph, often a brain parcel.
NIfTI	Neuroimaging file format commonly used for volumetric images.
Parcellation	Partitioning brain locations into regions.
ROI	Region of interest.
Vertex	A point in a surface mesh.
Voxel	A three-dimensional image element.

Further reading and official resources

For exact file conventions and command usage, prioritize the documentation shipped with your HCP release and the official Connectome Workbench documentation. Useful starting points:

- Connectome Workbench: <https://www.humanconnectome.org/software/connectome-workbench>
- HCP Young Adult study: <https://www.humanconnectome.org/study/hcp-young-adult>
- NIfTI format information: <https://nifti.nimh.nih.gov/>
- GIFTI specification and resources: <https://www.nitrc.org/projects/gifti/>

This guide is a learning reference, not a replacement for the metadata or processing documentation of a specific HCP file.